

Sample questions for the exam

Table 2 provides examples of exam questions.

Table 2. Examples of exam questions

1	What are Operational Technologies (OT)?	IT systems for power grids Information and communication technologies used for energy markets Information and communication technologies used for power grid monitoring and control
2	Substations communicate with the power system control centre via	Public communication networks The Internet Private communication networks
3	The wide area communication network is considered to be a	Trusted network Semi-trusted network Untrusted network
4	What security control is typically used to segregate the IT and OT systems at control centres for power grids?	Firewall Encryption Intrusion detection system
5	Cyber attacks on power grids are	Low impact, low frequency disturbances High impact, high frequency disturbances High impact, low frequency disturbances

6	What kind of facilities did Stuxnet malware target?	Nuclear Hydro power plants Windfarms
7	What was the impact of the Ukraine 2015 cyber attack?	Power outages affecting 225,000 customers Complete blackout affecting 500,000 customers Power system instability
8	How did attackers intrude in the utility IT system in the Ukraine 2015 cyber attack?	By using phishing e-mails By stealing credentials By using an infected USB
9	What is the impact of injecting spoofed IEC 61850 GOOSE traffic into the substation communication network for power grids?	Maliciously modify power generation Maliciously open circuit breakers to disconnect transmission lines Maliciously change voltage setpoints
10	What is the recommendation of IEC 62351-6 standard to secure the IEC 61850 protocols?	Use HMAC to check the integrity of the IEC 61850 data frames Use virtual private networks Use firewalls and antivirus in substations
11	The Internet backbone consists of a group of regional ISPs.	True False
12	Which of the following elements are placed at critical points within a system to protect vital assets?	Firewall IDS IPS All of the above
13	Identify all deployment requirements that should be considered when determining the locations and features of firewalls.	Write your answer here.

14	Describe the limitations of firewall inspection.	Write your answer here.
15	Describe and compare the key functions of IDS and IPS.	Write your answer here.